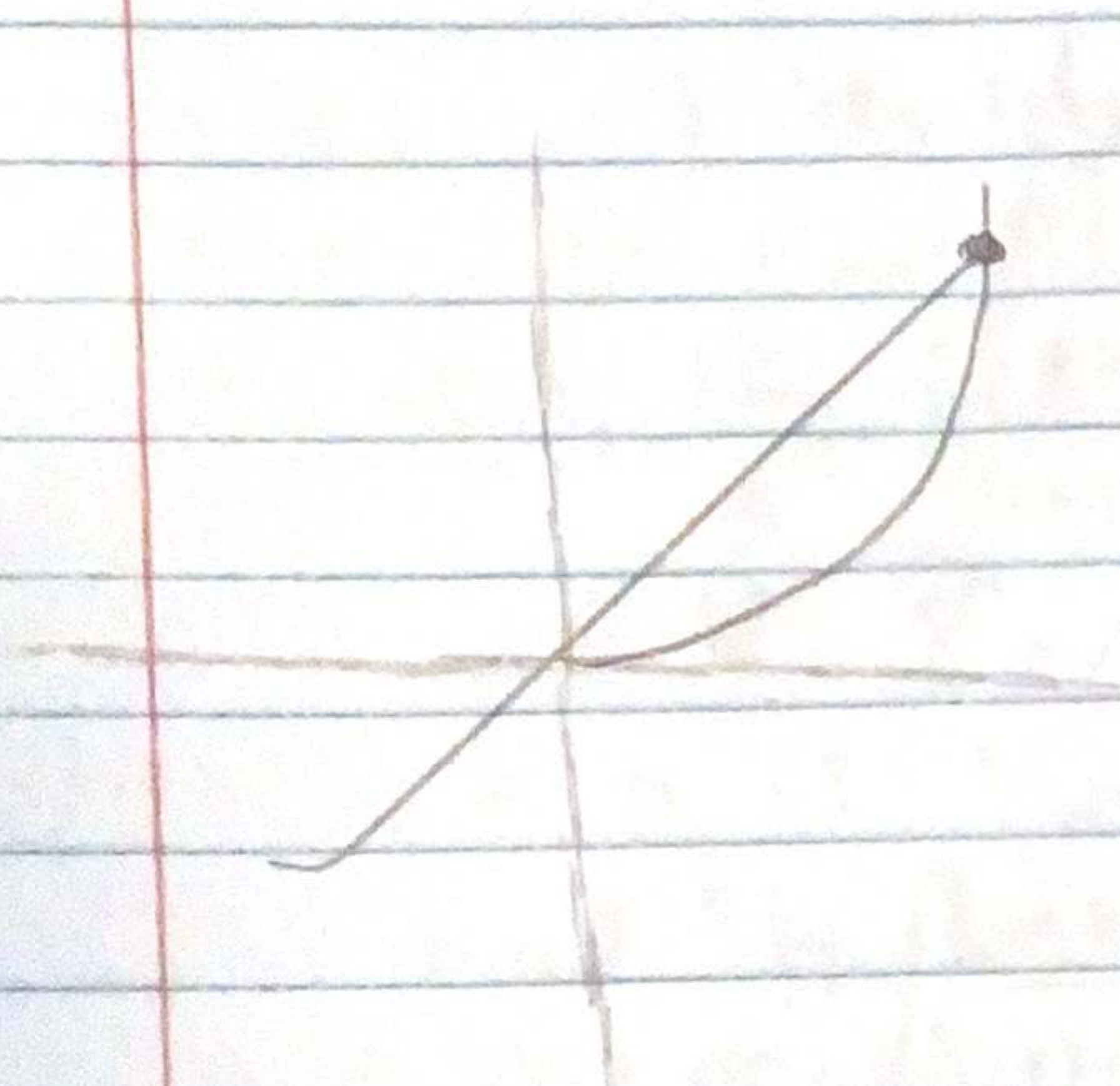


2.1 derivatives and rate of change

$$\frac{f(x) - f(a)}{x - a} = \frac{\text{difference in } y\text{-value}}{\text{difference in } x\text{-value}}$$



$$\frac{f(x) - f(a)}{x - a} = \text{rate of change}$$

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \text{instantaneous rate of change}$$

- Secant line touches the graph at two points
- tangent line touches the graph at one point

$$\underbrace{f'(a)}_{\text{Notation}} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$f'(a)$ mean the same as $\frac{df}{dx}$

$$\frac{df}{dx} = \frac{\text{change in } y}{\text{change in } x}$$

Example 1:

Given $f(x) = \frac{7-x}{2x+1}$

Find $f'(1)$

Sketch the equation of the tangent line

$$f(1) = \frac{7-1}{2(1)+1} = \frac{6}{3} = 2$$

$$\frac{f(x) - f(a)}{x - a}$$

$$f(1) = 2$$

$$\frac{\frac{7-x}{2x+1} - 2}{x-1} \rightarrow \left[\frac{7-x}{2x+1} - \frac{2(2x+1)}{2x+1} \right] \cdot \frac{1}{x-1}$$

$$\frac{7-x-4x-2}{2x+1} = \frac{5-5x}{2x+1}$$

$$\frac{-5(x-1)}{2x+1} \cdot \frac{1}{x-1} = \frac{-5}{2(1)+1} = \frac{-5}{3} \quad \boxed{f'(1) = \frac{-5}{3}}$$

(Slope)

Equation of tangent line

$$\text{Slope} = f'(1) = -5/3$$

$$\text{Point} = (1, 2)$$

x_1, y_1

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{-5}{3}(x - 1) \Rightarrow y - 2 = \frac{-5}{3}x + \frac{5}{3}$$

$$\boxed{y = \frac{-5}{3}x + \frac{11}{3}}$$

$$\frac{f(x) - f(a)}{x - a}$$

$$x - a$$

- $f(x)$ = is the equation you plug that in for $f(x)$ on example one we plugged that whole equation for $f(x)$

- $f(a) = a$ is your number that you are trying to find the derivative of plug that number into the equation use the answer as a replacement for " $f(a)$ "

- $(x - a) =$ for " a " you will just plug in the number you want to find the derivative of on the first example we wanted to find the derivative of " 1 " so " a " will be one

Equation of tangent line

the slope at $f'(1)$ is $-5/3$ and the point is $(1, 2)$ we got this point when we plugged 1 into the $f(x)$ function and it gave us 2

then just plug into the equation

difference quotient

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- this equation is used to find the derivative of the function by taking its limit as the change in x approach zero meaning when you take the limit of the difference quotient you will get a equation for the derivative

given $f(x) = \sqrt{2x+3}$ Find $f'(x) / \frac{df}{dx}$

$$\frac{\sqrt{2(x+h)+3} - \sqrt{2x+3}}{h} \cdot \frac{\sqrt{2x+h+3} + \sqrt{2x+3}}{\sqrt{2x+h+3} + \sqrt{2x+3}}$$

$$\frac{(\sqrt{2x+h+3})^2 - (\sqrt{2x+3})^2}{h(\sqrt{2x+h+3} + \sqrt{2x+3})} = \frac{2x+h+3 - 2x-3}{h(\sqrt{2x+h+3} + \sqrt{2x+3})}$$

$$\frac{2h}{h(\sqrt{2x+h+3} + \sqrt{2x+3})} = \frac{2}{\sqrt{2x+h+3} + \sqrt{2x+3}}$$

$$f'(x) = \frac{2}{2\sqrt{2x+3}}$$

• the difference quotient will give you a formula if you want to find the derivative of any number just plug it in

$$f(x) = \frac{x}{x+2}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)h}{h}$$

$$\frac{\frac{(x+h)}{(x+h)+2} - \frac{x}{x+2}}{h} \rightarrow \frac{\frac{(x+h)}{(x+h+2)} - \left(\frac{x}{x+2}\right)}{h}$$

$$\frac{(x+h)(x+2) - x(x+h+2)}{(x+h+2)(x+2)} \rightarrow \frac{x^2 + 2x + xh + 2h - x^2 - xh - 2x}{(x+h+2)(x+2)}$$

$$\frac{2h}{(x+h+2)(x+2)} \cdot \frac{1}{h} = \frac{2}{(x+2)^2}$$

$$f'(x) = \frac{2}{(x+2)^2}$$

Find the equation of the tangent line
(1, 0)

3.1 Derivatives

Power rule: $f(x) = x^n$ then $f'(x) = nx^{n-1}$

examples: $f(x) = x^2$ $f(x) = x^{10}$
 $f'(x) = 2x^1$ $f'(x) = 10x^9$

$f(x) = 3x^4$ $f(x) = 10x^5$
 $f'(x) = 12x^3$ $f'(x) = 50x^4$

- the Power rule consists of Bringing the exponent down and multiplying it with whatever is in front of x (if there is nothing there is always an invisible one) after bring it down subtract one from the exponent

- this only works for the numerator if you have a variable with an exponent on the denominator you need to bring it up

example: $f(x) = \frac{10}{x^2} \rightarrow 10x^{-2}$
 $f'(x) = -20x^{-3}$ or $-\frac{20}{x^3}$

$f(x) = \frac{1}{\sqrt{x}} \rightarrow (x)^{-1/2}$
 $f'(x) = -\frac{1}{2}x^{-3/2}$ or $-\frac{1}{2x^{3/2}}$

When you have a x^{-2} and you want to make it positive you can do this $\frac{1}{x^2}$ you can

also bring it up but it has to be x^{-2} for the square root $\sqrt{x}$ it can also be written as $(x)^{1/2}$ remember that $\sqrt[3]{x}$ the two is default unless it says otherwise

another example from the Power rule

if you had $\frac{11}{2t}$ the "t" would come up
not the both

$$\frac{11}{2t} = \frac{11}{2} t^{-1}$$

Constant rule: $y = c$ then $\frac{dy}{dx} = 0$

When there is a constant the answer for that constant is just a zero

example of constants: π , any number by itself, $4e^7$

$$\frac{dy}{dx} = 18x^5 + x^4 + y^2 \leftarrow \begin{array}{l} \text{There is a "y" the} \\ \text{question is looking for change} \\ \text{on "x" so it's a constant} \end{array}$$
$$90x^4 + 4x^3 + 0$$

the question is looking for "x" not "y"
so "y" is the constant

if the question said $\frac{dy}{d\pi}$ then the whole answer would be a constant because there is no " π " so the answer is zero

Constant multiple rule

$$y = c \cdot f(x) \text{ then } \frac{dy}{dx} = c \cdot \frac{df}{dx}$$

a number or a variable that is a constant being multiplied by something that is not a constant

example:

$2x^2$
Constant (pointing to 2)
Non-constant (pointing to x^2)

$$f'(x) = 2(x^2) \rightarrow 2(2x^{2-1}) \rightarrow 2(2x')$$

$$f'(x) = 4x$$

$$f(x) = \frac{10x^3}{7} + 4$$

$$\frac{df}{dx} = \frac{10}{7}x^3 + 4$$

$$\frac{df}{dx} = \frac{10}{7}(3x^2) + 0$$

$$\frac{df}{dx} = \frac{30}{7}x^2$$

exponential function

$$y = e^x \text{ then } \frac{dy}{dx} = e^x$$

if there is a exponential function with nothing but a simple x on top of it the you keep it the same will will get into exponential function and their input later

example:

→ the derivative would be the same

$$f(x) = 9e^x + 3x^2$$

$$\frac{df}{dx} = 9e^x + 6x$$

example #2:

→ The derivative would be the same

$$f(x) = \frac{1}{5}e^x - 10x^2$$

$$\frac{df}{dx} = \frac{1}{5}e^x - 20x$$

Product rule

$$y = f \cdot g \quad \text{then} \quad \frac{dy}{dx} = \frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

this rule can only be used for multiplication
keep in mind that there can be other rules
within the Product rule

example:

$$y = x^2 + e^x + xe^x$$

Power
rule

exponential rule

Product rule

first variable

second variable

$$F = x$$

$$g = e^x$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$\frac{df}{dx} = 1$$

$$\frac{dg}{dx} = e^x$$

$$(1)(e^x) + (x)(e^x)$$

$$e^x + xe^x$$

$$y = 2x + e^x + e^x + xe^x$$

$$\frac{dy}{dx} = 2x + 2e^x + xe^x$$

Quotient rule

$$y = \frac{f}{g} \text{ then } \frac{dy}{dx} = \frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2}$$

the rule can only be used for division
nothing else this rule should be used
when there are more than one term in
the denominator

Example: $\frac{x^2 + x^3 + \sqrt{x}}{x} = \frac{x^2}{x} + \frac{x^3}{x} + \frac{(x)^{1/2}}{x}$

When there is just one term you can
break it up in all of the numerator
then you can bring it up and make it
a negative or you can subtract

Example of when you should use the quotient
rule

Example #2:

$$\frac{3x+1}{x^2-x}$$

there are two term
in the denominator

top is f
bottom is g

$$f = 3x + 1$$

$$\frac{df}{dx} = 3$$

$$g = x^2 - x$$

$$\frac{dg}{dx} = 2x - 1$$

→

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2}$$

$$(3)(x^2 - x) - (3x - 1)(2x - 1)$$

$$3x^2 - 3x - 6x + 3x + 2x - 1$$

$$(x^2 - x)^2$$

$$(x^2 - x)^2$$

$$\frac{df}{dx} = \frac{-3x^2 - 4x - 1}{(x^2 - x)^2}$$

derivative of trig functions

$$\sin x = \frac{dy}{dx} = \cos x$$

$$\tan x = \frac{dy}{dx} = \sec^2 x$$

$$\sec x = \frac{dy}{dx} = \tan x \sec x$$

$$\cos x = \frac{dy}{dx} = -\sin x$$

$$\cot x = \frac{dy}{dx} = -\csc^2 x$$

$$\csc x = \frac{dy}{dx} = -\csc x \cot x$$

examples:

$$y = 3 \sin x + \cos x + \pi$$

$$\frac{dy}{dx} = 3 \cos x - \sin x + 0$$

More example on Next Pages



if you want you can treat this just as a normal term or you can apply Product rule you will get the same answer

$$y = \frac{2 \tan x}{3x}$$

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2}$$

$$f = 2 \tan x$$

$$g = 3x$$

$$g^2$$

$$\frac{df}{dx} = 2 \sec^2 x$$

$$\frac{dg}{dx} = 3$$

$$\frac{(2 \sec^2 x)(3x) - (2 \tan x)(3)}{(3x)^2}$$

$$\frac{dy}{dx} = \frac{6x \sec^2 x - 6 \tan x}{9x^2}$$

Chain rule

Chain rule can only be applied when the input is other than (x) this also includes " x^2 "

input other than " x "

$$\tan(x^3 + x^2)$$

take the derivative of the outer function this one is highlighted in green you will keep the input the way it is than you will multiply it all by the derivative of the input

$$\sec^2(x^3 + x^2) \cdot \text{derivative of input}$$

↑
input

↕

$$\sec^2(x^3 + x^2) \cdot (3x^2 + 2x)$$

example #1:

$$\csc(4x^2 - x^3)$$

input

all trig function have their own input

$$-\cot(4x^2 - x^3) \csc(4x^2 - x^3) \cdot \text{derivative of input}$$

$$\frac{dy}{dx}$$

$$-\cot(4x^2 - x^3) \csc(4x^2 - x^3) \cdot 8x - 3x^2$$

example #2:

$$y = \tan(2x)$$

$$\frac{dy}{dx} = \sec^2(2x) \cdot \text{derivative of input}$$

$$\frac{dy}{dx} = \sec^2(2x) \cdot 2$$

$$\boxed{\frac{dy}{dx} = 2 \sec^2(2x)}$$

example #3:

$$y = \csc(\sqrt[3]{x})$$

$$\frac{dy}{dx} = -\cot(\sqrt[3]{x}) \csc(\sqrt[3]{x}) \cdot \text{derivative of input}$$

$$\boxed{\frac{dy}{dx} = -\cot(\sqrt[3]{x}) \csc(\sqrt[3]{x}) \cdot \frac{1}{3x^{2/3}}}$$

implicit differentiation

the equation look like this for implicit differentiation

$$x^2 + 7y^2 = x + y$$

Notice how there are x 's and y 's on both sides of the equal sign. For implicit differentiation you will have to factor out the $\frac{dy}{dx}$.

$$x^2 + 7y^2 = x + y$$

Find derivative like any normal derivative but because there is a "y" multiply by $\frac{dy}{dx}$.

$$2x + 14y \cdot \frac{dy}{dx} = x + \frac{dy}{dx}$$

$$-14y \frac{dy}{dx} \quad -x$$

$$2x - x = -14y \frac{dy}{dx} + \frac{dy}{dx}$$

$$x = \frac{dy}{dx} (14y + 1)$$

$$\frac{x}{14y + 1} = \frac{dy}{dx}$$

- When there is something other than an "x" you will find the derivative like normal and multiply that "None x" with $\frac{dy}{dx}$. So later you can factor it out.

$$7x^2y^2 = x + e^y$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$f = 7x^2 \quad g = y^2$$

$$(14x)(y^2) + (7x^2)(2y \frac{dy}{dx})$$

$$\frac{df}{dx} = 14x \quad \frac{dg}{dx} = 2y \cdot \frac{dy}{dx}$$

$$14xy^2 + 14x^2y \frac{dy}{dx}$$

$$14xy^2 + 14x^2y \frac{dy}{dx} = 1 + e^y \cdot \frac{dy}{dx}$$

$$-14x^2y \frac{dy}{dx} \quad -1$$

$$14xy^2 - 1 = -14x^2y \frac{dy}{dx} + e^y \frac{dy}{dx}$$

$$14xy^2 - 1 = \frac{dy}{dx} (-14x^2y + e^y)$$

$$\boxed{\frac{14xy^2 - 1}{-14x^2y + e^y} = \frac{dy}{dx}}$$

$$\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$$

$$\sqrt{\frac{x}{y}} = \left(\frac{x}{y}\right)^{1/2} \rightarrow \frac{x^{1/2}}{y^{1/2}} \rightarrow \frac{\sqrt{x}}{\sqrt{y}}$$

$$\left(\frac{x}{y}\right)^{-1} = \frac{x^{-1}}{y^{-1}} \rightarrow \frac{\frac{1}{x}}{\frac{1}{y}} \rightarrow \frac{1}{x} \cdot \frac{y}{1} \rightarrow \frac{y}{x}$$

(Keep Change Flip)

$$\left(\frac{x}{y}\right)^{-1/2} = \frac{x^{-1/2}}{y^{-1/2}} \rightarrow \frac{\frac{1}{x^{1/2}}}{\frac{1}{y^{1/2}}} \rightarrow \frac{1}{x^{1/2}} \cdot \frac{y^{1/2}}{1} \rightarrow \frac{y^{1/2}}{x^{1/2}}$$

logarithmic functions

$$\ln(xy) = \ln(x) + \ln(y)$$

$$\ln(x/y) = \ln(x) - \ln(y)$$

$$\ln(x^n) = n \ln(x)$$

$$\ln(e) = 1$$

all of these are logarithmic properties
these are not derivatives

logarithmic and exponential derivatives

$$\ln(x) = \frac{dy}{dx} = \frac{1}{x}$$

$$\log_a x = \frac{dy}{dx} = \frac{1}{x \ln(a)}$$

example:

$$y = 5 \ln x + 7e^x$$

$$\frac{dy}{dx} = 5 \cdot \frac{1}{x} + 7e^x \rightarrow$$

$$\frac{dy}{dx} = \frac{5}{x} + 7e^x$$

examples:

(introducing "LN" to equations)

$$y = \frac{x^2}{\sqrt{x^3+1}} \rightarrow \ln(y) = \ln\left(\frac{x^2}{\sqrt{x^3+1}}\right)$$

$$\ln(y) = \ln(x^2) - \ln(x^3+1)^{1/2}$$

$$\ln(y) = 2\ln(x) - \frac{1}{2}\ln(x^3+1)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 2 \cdot \frac{1}{x} - \frac{1}{2} \cdot \frac{1}{x^3+1} \cdot \frac{dI}{dx}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{2}{x} - \frac{1}{2(x^3+1)} = 3x^2$$

$$y \cdot \frac{1}{y} \cdot \frac{dy}{dx} = \left(\frac{2}{x} - \frac{3x^2}{2x^3+2}\right) y \rightarrow \frac{dy}{dx} = \frac{x^2}{\sqrt{x^3+1}} \left(\frac{2}{x} - \frac{3x^2}{2x^3+2}\right)$$

you do not have to introduce "LN" to problems
you can apply the quotient rule if you want

- when you introduced "LN" you must introduce
it to both sides when that was done the
problem became a implicit differential

- you divided "y" to get $\frac{dy}{dx}$ by itself we
already know what "y" is so you need
to plug it in (y is the original question)

x both input

$$y = x^{\sin(x)}$$

(For this Problem you must introduce "LN" because there is an "x" in both inputs)

$$\ln(y) = \ln(x^{\sin(x)})$$

$$\ln(y) = \sin(x) \ln(x)$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \text{Product rule}$$

$$f = \sin(x)$$

$$g = \ln(x)$$

$$\frac{df}{dx} = \cos(x)$$

$$\frac{dg}{dx} = \frac{1}{x}$$

$$\left(\cos(x) \right) \left(\ln(x) \right) + \left(\sin(x) \right) \left(\frac{1}{x} \right)$$

$$\cos(x) \ln(x) + \frac{\sin(x)}{x}$$

$$y \cdot \frac{1}{y} \cdot \frac{dy}{dx} = \left(\cos(x) \ln(x) + \frac{\sin(x)}{x} \right)$$

$$\frac{dy}{dx} = y \left(\cos(x) \ln(x) + \frac{\sin(x)}{x} \right)$$

$$\frac{dy}{dx} = x^{\sin(x)} \left(\cos(x) \ln(x) + \frac{\sin(x)}{x} \right)$$

- we already know what "y" is so we will need to plug it into "y"

$$\log(xy) + x^2 - y = 0$$

$$\frac{1}{(xy)\log(10)} \cdot \frac{dI}{dx} + 2x - \frac{dy}{dx}$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$(1)(y) + (x)\left(\frac{dy}{dx}\right)$$

$$f = x$$

$$g = y$$

$$\frac{df}{dx} = 1$$

$$\frac{dg}{dx} = \frac{dy}{dx}$$

$$y + x \frac{dy}{dx}$$

$$\frac{1}{(xy)\log(10)} \cdot \left(y + x \frac{dy}{dx} \right) + 2x - \frac{dy}{dx} = 0$$

distribute

$$\frac{y}{(xy)\log(10)} + \frac{\frac{dy}{dx}}{(xy)\log(10)} + 2x - \frac{dy}{dx} = 0$$

$$\frac{\frac{dy}{dx}}{(xy)\log(10)} - \frac{dy}{dx} = -\frac{y}{(xy)\log(10)} - 2x$$

$$\log(10) = 1$$

$$\frac{dy}{dx} \left(\frac{1}{(xy)(1)} - 1 \right) = \frac{-y}{(xy)(1)} - 2x$$

$$\frac{dy}{dx} = \left[\frac{-y}{xy} - 2x \right] \cdot \frac{xy}{1 - xy} = \boxed{\frac{-y - 2x^2y}{1 - xy}}$$

$$\tan^{-1}(x) = \frac{1}{1+x^2}$$

$$y = \tan^{-1}(x^3 + 4)$$

$$\frac{dy}{dx} = \frac{1}{1+(x^3+4)^2} \cdot \frac{dI}{dx}$$

$$\frac{dy}{dx} = \frac{1}{1+(x^3+4)^2} \cdot 3x^2 \rightarrow \boxed{\frac{dy}{dx} = \frac{3x^2}{1+(x^3+4)^2}}$$

$$y = \tan^{-1}\left(\frac{1}{x}\right)$$

$$\frac{dy}{dx} = \frac{1}{1+\left(\frac{1}{x}\right)^2} \cdot \frac{dI}{dx} \left(\frac{1}{x} \rightarrow x^{-1}\right)$$

$$\frac{dy}{dx} = \frac{1}{1+\frac{1}{x^2}} \cdot -x^{-2}$$

$$\frac{dy}{dx} = \frac{-x^{-2}}{1+\frac{1}{x^2}} \rightarrow \frac{\boxed{-x^{-2}} \cdot \boxed{\frac{x^2}{1}}}{\boxed{1+\frac{1}{x^2}} \cdot \boxed{\frac{x^2}{1}}}$$

$$\boxed{\frac{dy}{dx} = \frac{-1}{x^2+1}}$$

$$x^{-2} \cdot x^2 = x^0 = 1$$